

Spanner 和 BigQuery：即時詐欺防護盾牌

來源：<https://codelabs.developers.google.com/next26/spanner-bigquery-graph?hl=zh-tw>

步驟數：8

瞭解 Spanner 和 BigQuery 如何為遊戲建立即時詐欺防禦機制。

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1. 簡介

歡迎來到《**Petverse**》後端，這是一款多人線上遊戲，玩家可以建立動物虛擬化身、探索世界，以及交易遊戲代幣。

最近，遊戲經濟面臨威脅。有錢的狗狗玩家帳戶中，大筆金額全數轉換成頂級鮭魚股票，我們懷疑有一群「羅賓漢貓」從富有的獵犬身上偷取食物，餵食大量貓咪。

在本程式碼研究室中，您將建構**即時詐欺防禦盾**，以偵測首腦和自動機器人集團。您將體驗 **Spanner** 中的作業資料如何與 **BigQuery** 中的分析資料匯聚，進而透過圖表 (GQL) 和關聯式查詢偵測及調查複雜的詐欺模式。

學習內容

- 使用 **BigQuery 持續查詢** 功能，偵測即時串流中的異常狀況，並將結果推送至 Spanner。
- 使用 **Spanner Graph** 追蹤財務關係，調查詐欺集團。
- 使用 **Spanner 向量搜尋** 找出相似的個人資料說明，找出機器人帳戶。
- 使用 **BigQuery Graph** 分析大量通訊記錄，找出協調圖。

軟硬體需求

- 已啟用計費功能的 Google Cloud 專案。
- 具備 SQL、終端機指令和 Python 的基本知識。
- 您可能需要 GitHub 帳戶 (程式碼代管於 GitHub)。

目標對象：中階開發人員、資料工程師和架構師。

預計總時長：45 到 60 分鐘。

費用預估：本程式碼研究室建立的資源費用應低於 \$5 美元。

2. 事前準備 / 設定

從做中學 免付費

立即取得 Google Cloud 抵免額，並完成這個實驗室。無須提供信用卡或付款方式。

info

從做中學 免付費

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啟用

💡 在 Next '26，您可以使用抵免額完成本程式碼研究室！掃描 Spanner 展位上的 QR code，或向展位工作人員索取抵免額代碼。

建立或選取 Google Cloud 專案

您必須擁有已啟用計費功能的 Google Cloud 專案，才能使用本實驗室所需的服務。

1. 在 [Google Cloud 控制台](#) 的專案選取器頁面中，選取或建立 Google Cloud 專案。
2. 確認 Cloud 專案已啟用計費功能。瞭解如何[確認是否已啟用計費功能](#)。
3. 在 [Cloud 控制台首頁](#) 中找出專案 ID

Google Cloud Super Project

Welcome

You're working in > ... > Super project

Project number: 123456789 Project ID: super-project-id

[Dashboard](#) [Cloud Hub](#)

+ Create an Agent with Agent Platform + Create a VM

+ Run a query in BigQuery + Deploy an application + Create a storage bucket

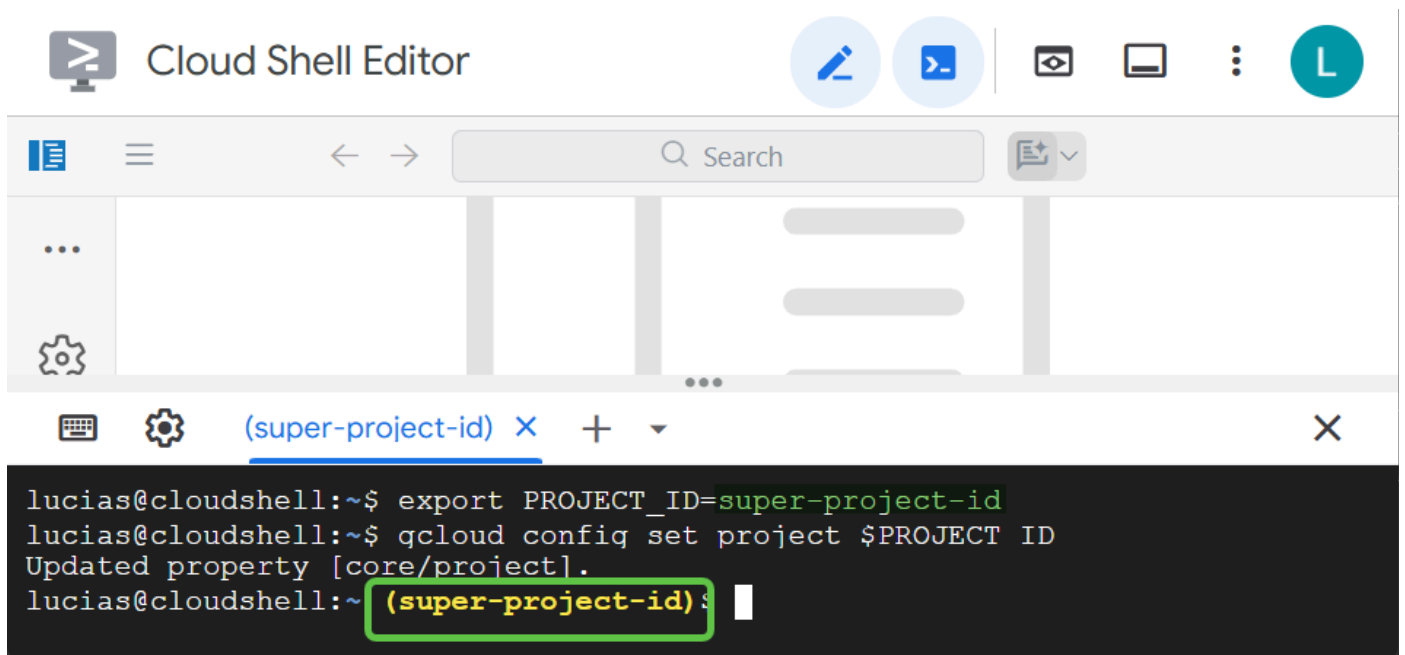
啟動 Cloud Shell

您將使用 [Google Cloud Shell](#) 做為執行環境。Cloud Shell 已預先安裝 `gcloud`、`git` 和其他必要工具。

1. 前往 [Google Cloud Shell](#)。
2. 如果出現提示訊息，請點選「授權」。
3. 在 Cloud Shell 終端機中設定環境變數，確保您在專案中工作：

將 `<YOUR_PROJECT_ID>` 替換為實際專案 ID。使用 Ctrl 或 Command + v 鍵貼到終端機。

```
export PROJECT_ID=<YOUR_PROJECT_ID>
gcloud config set project $PROJECT_ID
```



啟用必要的 API

執行下列指令，啟用 Spanner、BigQuery 和 Vertex AI 的 API：

```
gcloud services enable spanner.googleapis.com \
  bigquery.googleapis.com \
  aiplatform.googleapis.com \
  run.googleapis.com
```

複製存放區

複製包含 [應用程式程式碼](#) 和範例結構定義的存放區：

```
git clone https://github.com/GoogleCloudPlatform/cloud-spanner-samples.git
cd cloud-spanner-samples/spanner-bq-fraud-defense
```

3. 佈建基礎架構

您現在要在 BigQuery 中設定資料倉儲，並在 Spanner 中設定作業資料庫。

設定 BigQuery 資料集和連線

BigQuery 會分析遊戲的遙測串流。

在 Google Cloud Shell 的控制台中，執行下列指令，確認專案 ID 仍處於設定狀態：

將 <YOUR_PROJECT_ID> 替換為實際專案 ID。使用 Ctrl 或 Command + v 鍵貼到終端機。

```
export PROJECT_ID=<YOUR_PROJECT_ID>
gcloud config set project $PROJECT_ID
```

1. 建立 `game_analytics` BigQuery 資料集：

```
bq mk -d --location=US game_analytics
```

2. 建立連線，連結至儲存空間 bucket，並視需要連結至 Spanner：

```
bq mk --connection --location=US --project_id=$PROJECT_ID \
  --connection_type=CLOUD_RESOURCE unicorn-connection
```

3. 使用結構定義檔案，為 `GameplayTelemetry`、`AccountSignals`、`Players` 和 `ChatLogs` 資料表建立結構定義：

```
bq query --use_legacy_sql=false < bq_schema.sql
```

設定 Spanner

Google Cloud Spanner 會處理營運即時交易。在本實驗室中，我們會使用經濟實惠的 100 個處理單元 (PU) 執行個體。

1. 建立 Spanner 執行個體：

```
gcloud spanner instances create game-instance \
  --config=regional-us-central1 \
  --description="Game Instance" \
  --processing-units=100 \
  --edition=ENTERPRISE
```

2. 建立 Spanner 資料庫 `game-db`：

```
gcloud spanner databases create game-db --instance=game-instance
```

3. 使用應用程式資料表 (`Players` 、 `Transactions` 、 `AccountSignals`) 和屬性圖表 (`PlayerNetwork`) 更新 `Spanner` 結構定義：

```
gcloud spanner databases ddl update game-db --instance=game-instance --ddl-  
file=spanner_schema.sql
```

4. 為多模態向量搜尋建立 `AvatarSearchIndex`：

```
gcloud spanner databases ddl update game-db --instance=game-instance \  
--ddl="CREATE SEARCH INDEX AvatarSearchIndex ON Players(AvatarDescriptionTokens)"
```

將資料匯入 BigQuery

1. 將範例資料匯入 BigQuery：

```
bq load --source_format=AVRO game_analytics.GameplayTelemetry gs://sample-data-and-  
media/spanner-bq-fraud-heist/GameplayTelemetry  
bq load --source_format=AVRO game_analytics.AccountSignals gs://sample-data-and-  
media/spanner-bq-fraud-heist/AccountSignals  
bq load --source_format=AVRO game_analytics.Players gs://sample-data-and-media/spanner-bq-  
fraud-heist/Players  
bq load --source_format=AVRO game_analytics.ChatLogs gs://sample-data-and-media/spanner-bq-  
fraud-heist/ChatLogs
```

3. 前往 [BigQuery 控制台](#)，探索 `game_analytics` 資料集中的資料。

The screenshot displays the Google Cloud BigQuery console interface. On the left, a sidebar shows the project hierarchy: `super-project-id / Datasets / game_analytics / Tables / AccountSignals`. The main panel shows the `AccountSignals` table details. A notification indicates it is a partitioned table. Below the notification, tabs for `Schema`, `Details`, `Preview`, `Table Explorer`, `Preview`, `Insights`, `Lineage`, and `Data` are visible. The `Schema` tab is active, showing a table with the following fields:

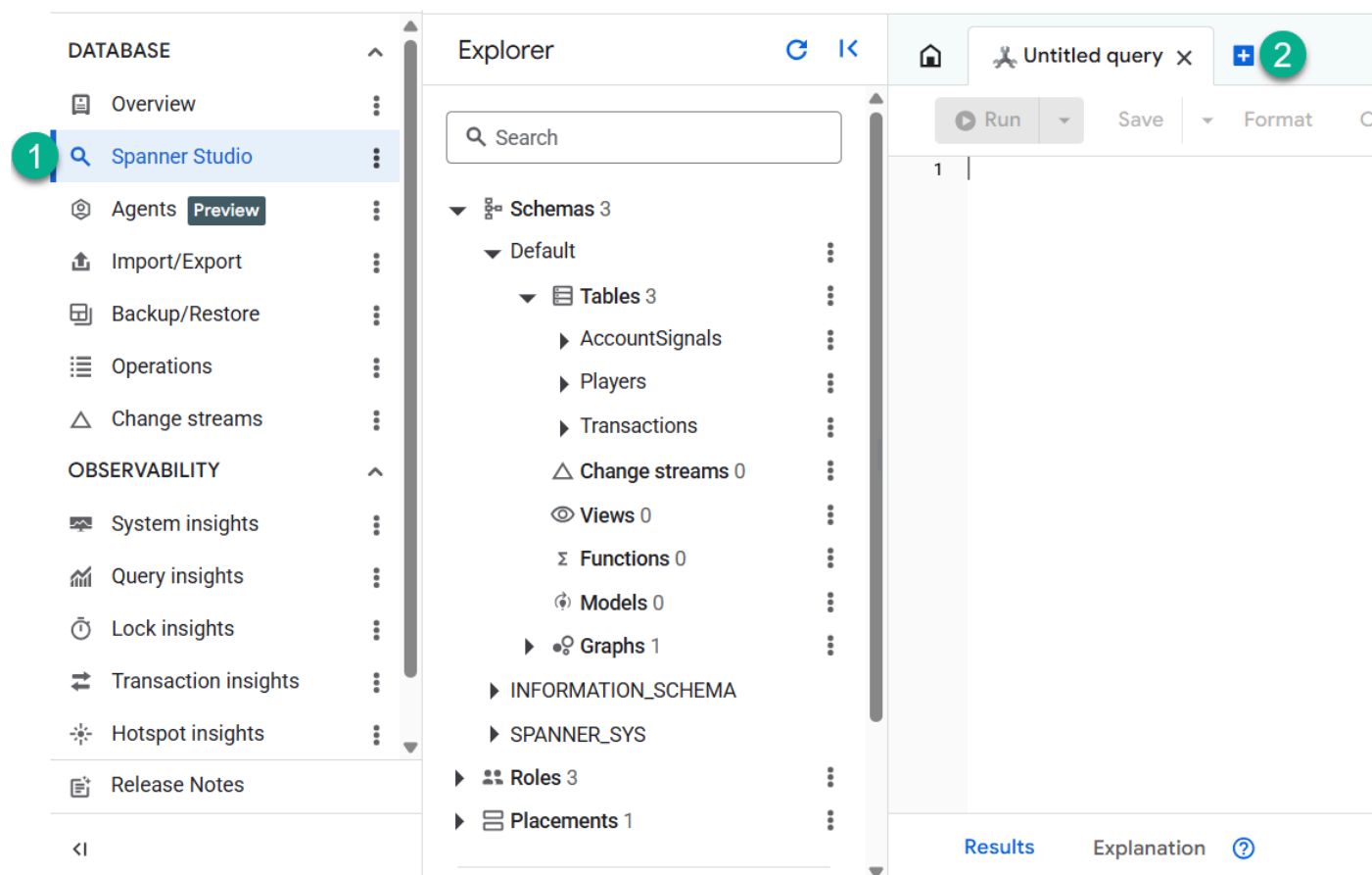
Field name	Type	Mode	Description	Key	Collation
SignalId	STRING	REQUIRED	-	-	-
PlayerId	STRING	REQUIRED	-	-	-
AlertType	STRING	REQUIRED	-	-	-
EventTime	TIMESTAMP	REQUIRED	-	-	-

At the bottom of the console, there are three buttons: `Edit schema`, `View row access policies`, and `Describe data`.

將資料匯入 Spanner

前往 [Spanner 控制台](#)，並探索 `game-db` 資料庫中的資料。

點選「Spanner Studio」，然後開啟「New Query (+)」。



將下列 `INSERT` 陳述式貼到查詢編輯器，然後點選「執行」：


```
-- Table: Players
INSERT INTO Players (
  PlayerId,
  Name,
  Species,
  Clan,
  AvatarDescription,
  ProfilePictureUrl,
  CreatedAt
) VALUES
  ('dc8cf07a-ac0f-48da-9f64-4f379492ble7', 'Pixel', 'Cat', 'CatClan', 'A heroic cat wearing a green tunic and a feathered cap', 'gs://sample-data-and-media/pixel_profile_booth.png', '2026-03-02T05:11:28.077335+00:00'),
  ('e82df4fb-0b6d-44dc-8609-70b41430af38', 'Rocky_1', 'Dog', 'DogClan', 'A robot dog with metal plating', 'gs://sample-data-and-media/wheaten_terrier_102.jpg', '2026-03-02T05:11:28.077374+00:00'),
  ('ea3afac7-54f0-4f68-8ed5-a5b6bd386c59', 'Whiskers_2', 'Cat', 'CatClan', 'A sneaky black cat hiding in the shadows', 'gs://sample-data-and-media/Bengal_100.jpg', '2026-03-02T05:11:28.077389+00:00'),
  ('f37d558b-fd0a-404c-a193-bc3a8a2edfba', 'Felix_3', 'Cat', 'CatClan', 'A cyber-punk cat with neon glasses', 'gs://sample-data-and-media/Abyssinian_1.jpg', '2026-03-02T05:11:28.077407+00:00'),
  ('f3687206-405e-43b6-afb0-8ca73eee5dd1', 'Luna_4', 'Cat', 'CatClan', 'A tabby cat with a red bandana', 'gs://sample-data-and-media/Bengal_100.jpg', '2026-03-02T05:11:28.077419+00:00'),
  ('82383e2d-d3a2-481d-b0cf-bfe165ed9bfd', 'Luna_5', 'Cat', 'CatClan', 'A fluffy persian cat with a golden collar', 'gs://sample-data-and-media/Abyssinian_114.jpg', '2026-03-02T05:11:28.077429+00:00'),
  ('755c7aff-e538-4681-9b90-4b870a42ac72', 'Buddy_6', 'Dog', 'DogClan', 'A tough bulldog with a spiked collar', 'gs://sample-data-and-media/yorkshire_terrier_101.jpg', '2026-03-02T05:11:28.077439+00:00'),
  ('8a034e84-26b3-4198-8ec9-3749b1f60537', 'Charlie_7', 'Dog', 'DogClan', 'A police german shepherd with a badge', 'gs://sample-data-and-media/wheaten_terrier_102.jpg', '2026-03-02T05:11:28.077462+00:00'),
  ('ca288a07-2bf8-46fa-a121-9bd0d0f44c64', 'Rocky_8', 'Dog', 'DogClan', 'A robot dog with metal plating', 'gs://sample-data-and-media/yorkshire_terrier_101.jpg', '2026-03-02T05:11:28.077474+00:00'),
  ('7b2881f0-289b-4ea4-9c0e-c1748249b70a', 'Bella_9', 'Dog', 'DogClan', 'A robot dog with metal plating', 'gs://sample-data-and-media/yorkshire_terrier_101.jpg', '2026-03-02T05:11:28.077484+00:00'),
  ('153f4022-a4ce-404a-8544-25d004fd34ad', 'Simba_10', 'Cat', 'CatClan', 'A tabby cat with a red bandana', 'gs://sample-data-and-media/Bengal_100.jpg', '2026-03-02T05:11:28.077494+00:00'),
  ('3fb82b8e-75a1-49fd-8691-a9dabb42bc4b', 'Charlie_11', 'Dog', 'DogClan', 'A police german shepherd with a badge', 'gs://sample-data-and-media/staffordshire_bull_terrier_116.jpg', '2026-03-02T05:11:28.077507+00:00'),
  ('e37b6dcf-7ccb-47d0-8b9d-0da2fa30ad09', 'Felix_12', 'Cat', 'CatClan', 'A cyber-punk cat with neon glasses', 'gs://sample-data-and-media/Bengal_105.jpg', '2026-03-02T05:11:28.077516+00:00'),
  ('0b395a7b-0673-4348-afd0-7cea9252629f', 'Bella_13', 'Dog', 'DogClan', 'A police german shepherd with a badge', 'gs://sample-data-and-media/yorkshire_terrier_101.jpg', '2026-03-02T05:11:28.077527+00:00'),
```

('64921a63-bea6-4a5c-9e49-7a817678c94f', 'Charlie_14', 'Dog', 'DogClan', 'A tough bulldog with a spiked collar', 'gs://sample-data-and-media/yorkshire_terrier_101.jpg', '2026-03-02T05:11:28.077537+00:00'),

('0d9040df-34a3-4e54-b660-a85e3c60a6fe', 'Felix_15', 'Cat', 'CatClan', 'A fluffy persian cat with a golden collar', 'gs://sample-data-and-media/Abyssinian_114.jpg', '2026-03-02T05:11:28.077546+00:00'),

('188c23a6-4c3b-4c25-9b40-9ec8ad7712a3', 'Max_16', 'Dog', 'DogClan', 'A fast greyhound wearing a racing vest', 'gs://sample-data-and-media/wheaten_terrier_102.jpg', '2026-03-02T05:11:28.077556+00:00'),

('a2eae9c9-dbff-4704-b908-74518d687e17', 'Rocky_17', 'Dog', 'DogClan', 'A tough bulldog with a spiked collar', 'gs://sample-data-and-media/shiba_inu_105.jpg', '2026-03-02T05:11:28.077565+00:00'),

('b59d9b9a-a169-43c1-832d-7e7d4acfa19c', 'Felix_18', 'Cat', 'CatClan', 'A sneaky black cat hiding in the shadows', 'gs://sample-data-and-media/Abyssinian_1.jpg', '2026-03-02T05:11:28.077581+00:00'),

('7019b91a-d908-4c93-8ed2-d1c60e518630', 'Nala_19', 'Cat', 'CatClan', 'A royal siamese cat wearing a crown', 'gs://sample-data-and-media/Abyssinian_1.jpg', '2026-03-02T05:11:28.077595+00:00'),

('1a972a71-a530-407a-b3f7-1dd9b2532cec', 'Luna_20', 'Cat', 'CatClan', 'A sneaky black cat hiding in the shadows', 'gs://sample-data-and-media/Bengal_105.jpg', '2026-03-02T05:11:28.077604+00:00'),

('3bd0489d-3f43-428d-be2d-0434f0e04ed8', 'Luna_21', 'Cat', 'CatClan', 'A cyber-punk cat with neon glasses', 'gs://sample-data-and-media/Bombay_104.jpg', '2026-03-02T05:11:28.077613+00:00'),

('28d7ab21-7ca7-435b-bdc4-5266a381ef89', 'Whiskers_22', 'Cat', 'CatClan', 'A fluffy persian cat with a golden collar', 'gs://sample-data-and-media/Bombay_104.jpg', '2026-03-02T05:11:28.077623+00:00'),

('87d880d4-c949-40e3-8d76-9307b45d57fe', 'Rocky_23', 'Dog', 'DogClan', 'A robot dog with metal plating', 'gs://sample-data-and-media/wheaten_terrier_102.jpg', '2026-03-02T05:11:28.077632+00:00'),

('b49a4523-83ff-4f59-8fca-54356928a18b', 'Nala_24', 'Cat', 'CatClan', 'A tabby cat with a red bandana', 'gs://sample-data-and-media/Bengal_105.jpg', '2026-03-02T05:11:28.077641+00:00'),

('1e049026-1231-4763-bc50-82cb51950fa2', 'Nala_25', 'Cat', 'CatClan', 'A fluffy persian cat with a golden collar', 'gs://sample-data-and-media/Abyssinian_1.jpg', '2026-03-02T05:11:28.077650+00:00'),

('e83caccd-e81e-454a-bf92-9d9a5a509e25', 'Simba_26', 'Cat', 'CatClan', 'A royal siamese cat wearing a crown', 'gs://sample-data-and-media/Bengal_105.jpg', '2026-03-02T05:11:28.077659+00:00'),

('45650b49-9701-4415-92a9-79073243d198', 'Simba_27', 'Cat', 'CatClan', 'A royal siamese cat wearing a crown', 'gs://sample-data-and-media/Bombay_104.jpg', '2026-03-02T05:11:28.077668+00:00'),

('2eb95b36-c9c3-4720-849b-fc240c9434da', 'Bella_28', 'Dog', 'DogClan', 'A robot dog with metal plating', 'gs://sample-data-and-media/samoyed_97.jpg', '2026-03-02T05:11:28.077677+00:00'),

('491943d2-8aa3-4484-9d6d-afcab7ec579a', 'Buddy_29', 'Dog', 'DogClan', 'A robot dog with metal plating', 'gs://sample-data-and-media/wheaten_terrier_102.jpg', '2026-03-02T05:11:28.077687+00:00'),

('004ac8b2-a37d-42c8-aba3-35b204f596f0', 'Max_30', 'Dog', 'DogClan', 'A police german shepherd with a badge', 'gs://sample-data-and-media/wheaten_terrier_102.jpg', '2026-03-02T05:11:28.077697+00:00'),

('5cfla55e-7904-4937-a4bd-bf3d1fclc839', 'Whiskers_31', 'Cat', 'CatClan', 'A royal siamese

cat wearing a crown', 'gs://sample-data-and-media/Bengal_100.jpg', '2026-03-02T05:11:28.077706+00:00'),

('4736fc20-1c22-49c2-af87-c35802302507', 'Luna_32', 'Cat', 'CatClan', 'A fluffy persian cat with a golden collar', 'gs://sample-data-and-media/Bengal_100.jpg', '2026-03-02T05:11:28.077715+00:00'),

('d98fc7ff-dd92-4723-994c-6267ef951bcc', 'Buddy_33', 'Dog', 'DogClan', 'A police german shepherd with a badge', 'gs://sample-data-and-media/staffordshire_bull_terrier_116.jpg', '2026-03-02T05:11:28.077797+00:00'),

('3de80488-e1b6-4908-a6ec-9c51f46f43b9', 'Luna_34', 'Cat', 'CatClan', 'A sneaky black cat hiding in the shadows', 'gs://sample-data-and-media/Bombay_104.jpg', '2026-03-02T05:11:28.077821+00:00'),

('710128e0-4e1d-479c-acbc-497bbc5bc802', 'Bella_35', 'Dog', 'DogClan', 'A tough bulldog with a spiked collar', 'gs://sample-data-and-media/scottish_terrier_108.jpg', '2026-03-02T05:11:28.077836+00:00'),

('7e5d416f-f42c-410e-8e2a-19be0cdfc1e5', 'Simba_36', 'Cat', 'CatClan', 'A royal siamese cat wearing a crown', 'gs://sample-data-and-media/Abyssinian_114.jpg', '2026-03-02T05:11:28.077850+00:00'),

('b5af580c-4998-4f68-b52f-5c5b6a9d0aab', 'Buddy_37', 'Dog', 'DogClan', 'A robot dog with metal plating', 'gs://sample-data-and-media/scottish_terrier_108.jpg', '2026-03-02T05:11:28.077864+00:00'),

('145cc805-810a-4320-864d-ba6b5c6fbc33', 'Max_38', 'Dog', 'DogClan', 'A tough bulldog with a spiked collar', 'gs://sample-data-and-media/staffordshire_bull_terrier_116.jpg', '2026-03-02T05:11:28.077884+00:00'),

('34fa84c8-3356-4ed6-9ac5-967cd2baabac', 'Whiskers_39', 'Cat', 'CatClan', 'A royal siamese cat wearing a crown', 'gs://sample-data-and-media/Bengal_105.jpg', '2026-03-02T05:11:28.077894+00:00'),

('a50c967b-2e89-448d-be47-4df628c51572', 'Luna_40', 'Cat', 'CatClan', 'A royal siamese cat wearing a crown', 'gs://sample-data-and-media/Bengal_105.jpg', '2026-03-02T05:11:28.077906+00:00'),

('64aeeb22-42db-46bf-85bd-b21b135c9803', 'Simba_41', 'Cat', 'CatClan', 'A cyber-punk cat with neon glasses', 'gs://sample-data-and-media/Bombay_104.jpg', '2026-03-02T05:11:28.077915+00:00'),

('dec825d6-91f4-4f8f-b44b-74cf6b4b78f6', 'Felix_42', 'Cat', 'CatClan', 'A sneaky black cat hiding in the shadows', 'gs://sample-data-and-media/Bengal_100.jpg', '2026-03-02T05:11:28.077924+00:00'),

('d9380e8d-8949-4849-af57-7a91a1a2b953', 'Bella_43', 'Dog', 'DogClan', 'A police german shepherd with a badge', 'gs://sample-data-and-media/staffordshire_bull_terrier_116.jpg', '2026-03-02T05:11:28.077933+00:00'),

('88a425a0-b499-4825-8a16-3d33b85feec9', 'Whiskers_44', 'Cat', 'CatClan', 'A fluffy persian cat with a golden collar', 'gs://sample-data-and-media/Abyssinian_114.jpg', '2026-03-02T05:11:28.077942+00:00'),

('e17ae312-37c6-4a66-9172-2d20d8528033', 'Simba_45', 'Cat', 'CatClan', 'A cyber-punk cat with neon glasses', 'gs://sample-data-and-media/Abyssinian_1.jpg', '2026-03-02T05:11:28.077958+00:00'),

('ae82282e-d380-4500-99e0-2dc4e276c0cf', 'Nala_46', 'Cat', 'CatClan', 'A cyber-punk cat with neon glasses', 'gs://sample-data-and-media/Bombay_104.jpg', '2026-03-02T05:11:28.077967+00:00'),

('9ce13d41-dede-4658-a7e4-7bc00372d65c', 'Simba_47', 'Cat', 'CatClan', 'A cyber-punk cat with neon glasses', 'gs://sample-data-and-media/Bengal_105.jpg', '2026-03-02T05:11:28.077976+00:00'),

('8db26b5c-2c09-4578-89bf-f9afaf765e36', 'Whiskers_48', 'Cat', 'CatClan', 'A fluffy persian cat with a golden collar', 'gs://sample-data-and-media/Bengal_100.jpg', '2026-03-

02T05:11:28.077989+00:00'),
('eb113965-6ee6-459e-ba0f-fb6cb1d0ae34', 'Luna_49', 'Cat', 'CatClan', 'A tabby cat with a red bandana', 'gs://sample-data-and-media/Bombay_104.jpg', '2026-03-02T05:11:28.078002+00:00'),
('3f064471-24a0-47cd-8927-925cb3e5df7f', 'Felix_50', 'Cat', 'CatClan', 'A tabby cat with a red bandana', 'gs://sample-data-and-media/Bengal_100.jpg', '2026-03-02T05:11:28.078012+00:00'),
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('7e9bbec4-e2aa-4d67-b1c0-35f7db1d0dae', 'Buddy_52', 'Dog', 'DogClan', 'A fast greyhound wearing a racing vest', 'gs://sample-data-and-media/yorkshire_terrier_101.jpg', '2026-03-02T05:11:28.078030+00:00'),
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('4e9dfce6-555b-434a-81b6-237c61b9b530', 'Merry_Cat_0', 'Cat', 'RobinHoods', 'A sneaky rebel cat wearing a green tunic and a feathered cap', 'gs://sample-data-and-media/Bombay_104.jpg', '2026-03-02T05:11:28.078112+00:00'),
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('78d66ff4-0519-4157-9d95-76c2900ba7f9', 'Merry_Cat_4', 'Cat', 'RobinHoods', 'A sneaky rebel cat wearing a green tunic and a feathered cap', 'gs://sample-data-and-media/Abyssinian_114.jpg', '2026-03-02T05:11:28.078301+00:00'),
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media/Bombay_104.jpg', '2026-03-02T05:11:28.078508+00:00'),
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-- Table: Transactions

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    SenderId,
    ReceiverId,
    Amount,
    Timestamp,
    IsSuspicious

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) VALUES

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-- Table: AccountSignals

-- This is pushed by BigQuery through a continuous query if it is configured

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INSERT INTO AccountSignals (SignalId, PlayerId, AlertType, EventTime) VALUES ('50b64a6a-2e8f-4a0b-9742-c7e180949e82', '4e9dfce6-555b-434a-81b6-237c61b9b530', 'SUSPICIOUS_MOVEMENT', '2026-03-02T05:11:28.078156+00:00');
INSERT INTO AccountSignals (SignalId, PlayerId, AlertType, EventTime) VALUES ('d2659e80-bce0-46b0-9570-6bb8b3c99d34', '686337b0-304e-4270-89c3-d015f9039294', 'SUSPICIOUS_MOVEMENT', '2026-03-02T05:11:28.078198+00:00');
INSERT INTO AccountSignals (SignalId, PlayerId, AlertType, EventTime) VALUES ('10504018-b7a4-47a4-89b2-950fa492bbd4', '1627bc18-c42e-4599-b5f1-6f3d52669edb', 'SUSPICIOUS_MOVEMENT', '2026-03-02T05:11:28.078244+00:00');
INSERT INTO AccountSignals (SignalId, PlayerId, AlertType, EventTime) VALUES ('ccbba78a-0f45-49b9-a58b-21777de250cf', '215b3d04-402a-4ac2-83ed-1edb9a421691', 'SUSPICIOUS_MOVEMENT', '2026-03-02T05:11:28.078293+00:00');
INSERT INTO AccountSignals (SignalId, PlayerId, AlertType, EventTime) VALUES ('affa841d-0f9a-4b0f-ad6d-eb3e180263fe', '78d66ff4-0519-4157-9d95-76c2900ba7f9', 'SUSPICIOUS_MOVEMENT', '2026-03-02T05:11:28.078337+00:00');
INSERT INTO AccountSignals (SignalId, PlayerId, AlertType, EventTime) VALUES ('1fbdd097-b431-4a21-9a5c-alf53db6d754', '822a8aae-57bd-460e-9643-69815990bec8', 'SUSPICIOUS_MOVEMENT', '2026-03-02T05:11:28.078383+00:00');
INSERT INTO AccountSignals (SignalId, PlayerId, AlertType, EventTime) VALUES ('0fc87891-dd4b-494b-8c47-f6ccc2c592c7', '9f223b65-41a3-495e-940a-5a07d2ba4ba7', 'SUSPICIOUS_MOVEMENT', '2026-03-02T05:11:28.078420+00:00');
INSERT INTO AccountSignals (SignalId, PlayerId, AlertType, EventTime) VALUES ('cc117869-7a6b-4e66-89bf-e3148ac8c6ea', '3910b199-caf7-4822-8346-1ba9bd750c8e', 'SUSPICIOUS_MOVEMENT', '2026-03-02T05:11:28.078456+00:00');
INSERT INTO AccountSignals (SignalId, PlayerId, AlertType, EventTime) VALUES ('c85a78c6-3cc8-4262-8fd2-2fb9661a1b96', '78c129b6-21e5-40c6-b9cf-d25ad5193e68', 'SUSPICIOUS_MOVEMENT', '2026-03-02T05:11:28.078499+00:00');
INSERT INTO AccountSignals (SignalId, PlayerId, AlertType, EventTime) VALUES ('ccdb20f8-cd05-42ae-b942-cb41396ec27d', 'eba84e9d-bd10-466e-8fcd-899c0a868149', 'SUSPICIOUS_MOVEMENT', '2026-03-02T05:11:28.078537+00:00');
INSERT INTO AccountSignals (SignalId, PlayerId, AlertType, EventTime) VALUES ('721511f1-ebc4-45dc-adb5-9dafc5426bc3', '37ale15c-a38c-4763-a2bd-042741bce012', 'SUSPICIOUS_MOVEMENT', '2026-03-02T05:11:28.078579+00:00');
INSERT INTO AccountSignals (SignalId, PlayerId, AlertType, EventTime) VALUES ('4d8b0f0d-211a-4537-8c79-c2fd815812f3', 'c5337857-3b58-4b0c-8814-4585dcbf765f', 'SUSPICIOUS_MOVEMENT', '2026-03-02T05:11:28.078627+00:00');
```

這項作業會在幾分鐘內完成。完成後即可預覽資料。

All instances

Instance

game-instance: Overview

Google Standard SQL Database

game-db: Spanner Studio

Open in Dataplex

Learn

Explorer

Search

Schemas 3

- Default
 - Tables 3
 - AccountSignals
 - Columns 4
 - Keys 2
 - Indexes 0
 - Players
 - Transactions
 - Change streams 0
 - Views 0
 - Functions 0
 - Models 0
 - Graphs 1

1

2

Run

Save

Format

Clear

Documentation

Valid

1

2

SELECT

*

FROM

'AccountSignals'

LIMIT

10;

Explanation

Export

Preview data

Insert data

Drop table

Equivalent DDL

Open in Dataplex

	PlayerId	AlertType	EventTime	
	"b-494b-8c47-f6ccc2c592c7"	"9f223b65-41a3-495e-940a-5a07d2ba4ba7"	"SUSPICIOUS_MOVEMENT"	2026-03-02T05:11:28.07842Z
	"4-47a4-89b2-950fa492bbd4"	"1627bc18-c42e-4599-b5f1-6f3d52669edb"	"SUSPICIOUS_MOVEMENT"	2026-03-02T05:11:28.078244Z
	"1-4a21-9a5c-a1f53db6d754"	"822a8aae-57bd-460e-9643-69815990bec8"	"SUSPICIOUS_MOVEMENT"	2026-03-02T05:11:28.078383Z
	"4d8b0f0d-211a-4537-8c79-c2fd815812f3"	"c5337857-3b58-4b0c-8814-4585dcbf765f"	"SUSPICIOUS_MOVEMENT"	2026-03-02T05:11:28.078627Z
	"50b64a6a-2e8f-4a0b-9742-c7e180949e82"	"4e9dfce6-555b-434a-81b6-237c61b9b530"	"SUSPICIOUS_MOVEMENT"	2026-03-02T05:11:28.078156Z
	"721511f1-ebc4-45dc-adb5-9dafc5426bc3"	"37a1e15c-a38c-4763-a2bd-042741bce012"	"SUSPICIOUS_MOVEMENT"	2026-03-02T05:11:28.078579Z
	"affa841d-0f9a-4b0f-ad6d-eb3e180263fe"	"78d66ff4-0519-4157-9d95-76c2900ba7f9"	"SUSPICIOUS_MOVEMENT"	2026-03-02T05:11:28.078337Z
	"c85a78c6-3cc8-4262-8fd2-2fb9661a1b96"	"78c129b6-21e5-40c6-b9cf-d25ad5193e68"	"SUSPICIOUS_MOVEMENT"	2026-03-02T05:11:28.078499Z

步驟 4 · 約 1 分鐘

4. Watchdog (BigQuery 持續查詢和 Spanner 同步)

我們的第一道防線是串流至 BigQuery 的遙測資料。我們想監控可疑移動 (例如不可能的距離)，並即時將快訊推送回 Spanner。

在實際情況中，您會使用 [BigQuery 持續查詢](#) 和反向 ETL 串流這項資料。不過，這需要 [預訂 ENTERPRISE 以上版本](#)。

如果預訂時間可供預約，指令會如下所示。如果您尚未設定預訂，則不需要將這項資訊複製到控制台中：

```
EXPORT DATA
  OPTIONS (
    uri = 'https://spanner.googleapis.com/projects/<YOUR_PROJECT_ID>/instances/game-
instance/databases/game-db',
    format='CLOUD_SPANNER',
    spanner_options="{ \"table\": \"AccountSignals\" }"
  ) AS
SELECT
  GENERATE_UUID() as SignalId,
  PlayerId,
  'SUSPICIOUS_MOVEMENT' as AlertType,
  CURRENT_TIMESTAMP() as EventTime
FROM `game_analytics.GameplayTelemetry`
WHERE
  EventType = 'player_move'
  AND (LocationX > 1000 OR LocationY > 1000);
```

如要建立持續查詢，請在 BigQuery 控制台的 SQL 工作區中，依序點選「更多」>「建立持續查詢」。

這項查詢會做為反向 ETL 引擎，確保交易系統 (Spanner) 能立即得知分析系統 (BigQuery) 偵測到的異常狀況。

在本實驗室中，我們在 Spanner 中人為插入了一些交易。

步驟 5 · 約 10 分鐘

5. 多模態偵探 (Spanner Graph 和向量搜尋)

現在 Spanner 具有「高風險」信號，您可以調查詐欺集團。我們會使用 **Spanner Graph**，以視覺化方式呈現金融網路，並找出主謀。

在 **Spanner Studio** 中執行這些查詢

圖表：找出幕後主使者

這項查詢會追蹤金融交易網路，受害者會將款項轉給竊賊，竊賊再轉給主節點。並依首領分組，加總戰利品。

複製到 Spanner Studio，然後點選「Run」（執行）。

```
GRAPH PlayerNetwork
MATCH (victim)-[:Transfers]->(thief)-[t:Transfers]->(boss)
RETURN
  boss.Name AS RingLeader, COUNT(t) AS TributesReceived,
  SUM(t.Amount) AS TotalLoot
GROUP BY RingLeader
ORDER BY TotalLoot DESC
LIMIT 5;
```

「Pixel」應會顯示為獲得最多致敬的收件者。

多信號調查

現在，我們將 Graph 結果與先前從 BigQuery 推送的即時行為信號合併。我們想找出將款項匯給「Pixel」且因可疑活動而遭到標記的玩家。

```
SELECT DISTINCT
  p.Name,
  s.AlertType as BQ_Signal,
  s.EventTime as SignalTime
FROM GRAPH_TABLE (
  PlayerNetwork
  MATCH (associate:Players)-[:Transfers]->(boss:Players)
  WHERE boss.Name = 'Pixel'
  RETURN DISTINCT associate.Name
) as g
JOIN Players p
  ON p.Name = g.Name
JOIN AccountSignals s
  ON p.PlayerId = s.PlayerId
ORDER BY s.EventTime DESC;
```

向量搜尋：找出機器人帳戶

這些是真人玩家還是有組織的機器人集團？使用**向量搜尋**功能，找出個人資料說明與「Pixel」極為相似的帳戶。

```
SELECT
  Name, AvatarDescription,
  COSINE_DISTANCE(AvatarEmbedding, (SELECT AvatarEmbedding FROM Players WHERE Name =
'Pixel')) as Similarity
FROM Players
WHERE Name != 'Pixel'
ORDER BY Similarity ASC
LIMIT 5;
```

相似度分數越低，表示越接近「Pixel」的向量。如果描述內容相似，很可能就是機器人！

我們也可以在 `MATCH` 子句中套用純量函式：

```
GRAPH PlayerNetwork
MATCH (associate:Players)-[:Transfers]->(boss:Players)
WHERE boss.Name = 'Pixel'
ORDER BY (
  COSINE_DISTANCE(associate.AvatarEmbedding, (SELECT AvatarEmbedding FROM Players WHERE Name
= 'Pixel'))
) ASC
RETURN DISTINCT associate.Name
```

步驟 6 · 約 8 分鐘

6. 揭開劇情 (BigQuery 屬性圖和 GCS 整合)

我們已逮捕首腦，但需要瞭解他們如何協調這場大規模的「鮪魚搶劫案」。我們將使用 **BigQuery 屬性圖** 查詢遊戲聊天記錄，追蹤 BigQuery 中的通訊模式。

在 **BigQuery Studio** 中執行下列查詢：

BigQuery 屬性圖

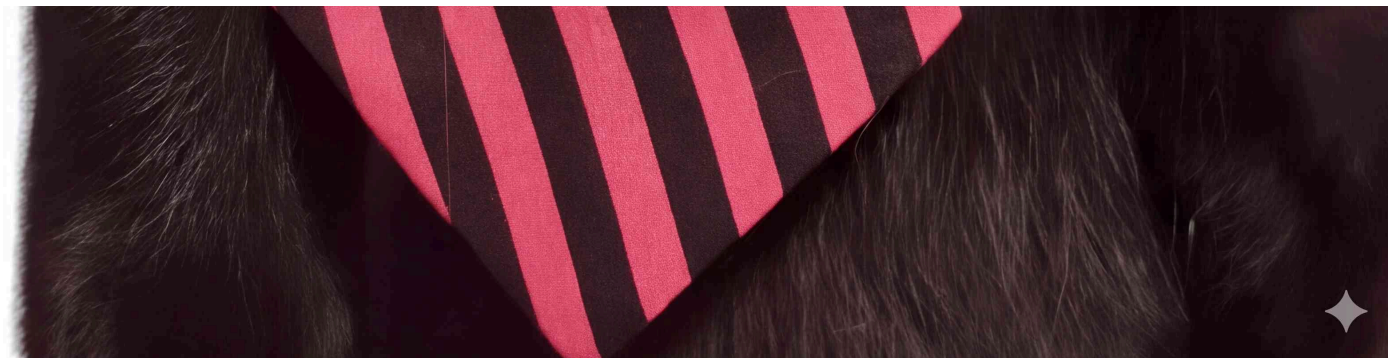
追蹤「Pixel」和其他播放器之間的通訊：

```
GRAPH game_analytics.CatChatNetwork
MATCH (p1:Players)-[c:Communicates]->(p2:Players)
WHERE p1.Name = 'Pixel' OR p2.Name = 'Pixel'
RETURN
  p1.Name AS Sender,
  p2.Name AS Receiver,
  c.Message,
  -- Resolving structured metadata from ObjectRef
  p1.ProfilePictureUrl.uri AS SenderProfilePic
ORDER BY Message DESC;
```

觀察「Operation Fishbowl is a go」(魚缸行動開始) 和「Diverting funds to the central tuna reserve」(將資金轉移至中央鮪魚儲備) 等訊息。您可以瞭解如何使用 BigQuery 屬性圖表，分析以非結構化資料 (使用 `ProfilePictureUrl.uri` 的 GCS 圖片參照) 擴增的通訊內容。

如果您下載圖片或點選結果中的已驗證 **GCS 連結**，就會看到球員的圖片：





這項分析查詢會進一步比較詐騙集團成員之間的對話模式和圖片。

如要執行這項操作，請先宣告多模態模型，為儲存在 Cloud Storage bucket 中的個人資料相片生成嵌入內容。這個模型會透過初始設定中建立的連線進行連線，因此您也會將 Vertex AI 使用者權限授予附加至該連線的技術使用者。

將 `<>` 替換為專案 ID。

```
GRANT `roles/aiplatform.user`  
ON PROJECT `<<PROJECT_ID>>`  
TO "connection:<<PROJECT_ID>>.us.unicorn-connection";
```

現在可以建立連線了。

```
CREATE OR REPLACE MODEL `game_analytics.multimodal_model`  
  REMOTE WITH CONNECTION `us.unicorn-connection`  
  OPTIONS (ENDPOINT = 'multimodalembembedding@001');
```

如果因權限錯誤而失敗 (即「The bqcx-12345745345345@gcp-sa-bigquery-condel.iam.gserviceaccount.com does not have the permission to access or use the endpoint...」)，請稍候幾分鐘，讓權限生效，然後再試一次。

下列查詢會使用 `AI.GENERATE_EMBEDDING` 函式，逐一處理儲存空間 bucket 中的圖片並建立嵌入。接著使用 `COSINE_DISTANCE` 比較這些嵌入內容，深入瞭解對話記錄和個人資料相片的相似程度。

```

-- BigQuery Property Graph: Tracing communication patterns in chat logs
-- AND calculating distance between auto-embedded chat message and profile picture
-- BigQuery Property Graph: Tracing communication patterns
-- AND identifying similarity AMONG the fraudsters themselves
WITH GraphResults AS (
  SELECT *
  FROM GRAPH_TABLE(
    game_analytics.CatChatNetwork
    MATCH (p1:Players)-[c:Communicates]->(p2:Players)
    WHERE p1.Name = 'Pixel' OR p2.Name = 'Pixel'
    RETURN
      p1.Name AS Sender,
      c.Message,
      p1.ProfilePictureUrl.uri AS SenderProfilePic,
      c.MessageEmbedding.result AS MessageEmbedding
  )
),
UniquePics AS (
  SELECT DISTINCT SenderProfilePic AS uri FROM GraphResults
),
PicEmbeddings AS (
  SELECT embedding, uri
  FROM AI.GENERATE_EMBEDDING(
    MODEL game_analytics.multimodal_model,
    (
      SELECT OBJ.MAKE_REF(uri, 'us.unicorn-connection') as content, uri
      FROM UniquePics
    )
  )
),
CatData AS (
  -- Distinct list of players (excluding Pixel) with their embeddings and HTTPS Pic URLs
  SELECT DISTINCT
    g.Sender,
    g.MessageEmbedding,
    g.Message,
    p.embedding AS PicEmbedding,
    REPLACE( g.SenderProfilePic, 'gs://sample-data-and-media/spanner-bq-fraud-
    heist/profile_pics/', 'https://storage.mtls.cloud.google.com/sample-data-and-media/spanner-
    bq-fraud-heist/profile_pics/') AS SenderProfilePic
  FROM GraphResults g
  LEFT JOIN PicEmbeddings p ON g.SenderProfilePic = p.uri
  WHERE g.Sender != 'Pixel'
    AND g.MessageEmbedding IS NOT NULL
    AND p.embedding IS NOT NULL
)
SELECT
  c1.Sender AS Fraudster_A,
  c2.Sender AS Fraudster_B,
  c1.SenderProfilePic AS Pic_A,
  c2.SenderProfilePic AS Pic_B,
  c1.Message,

```

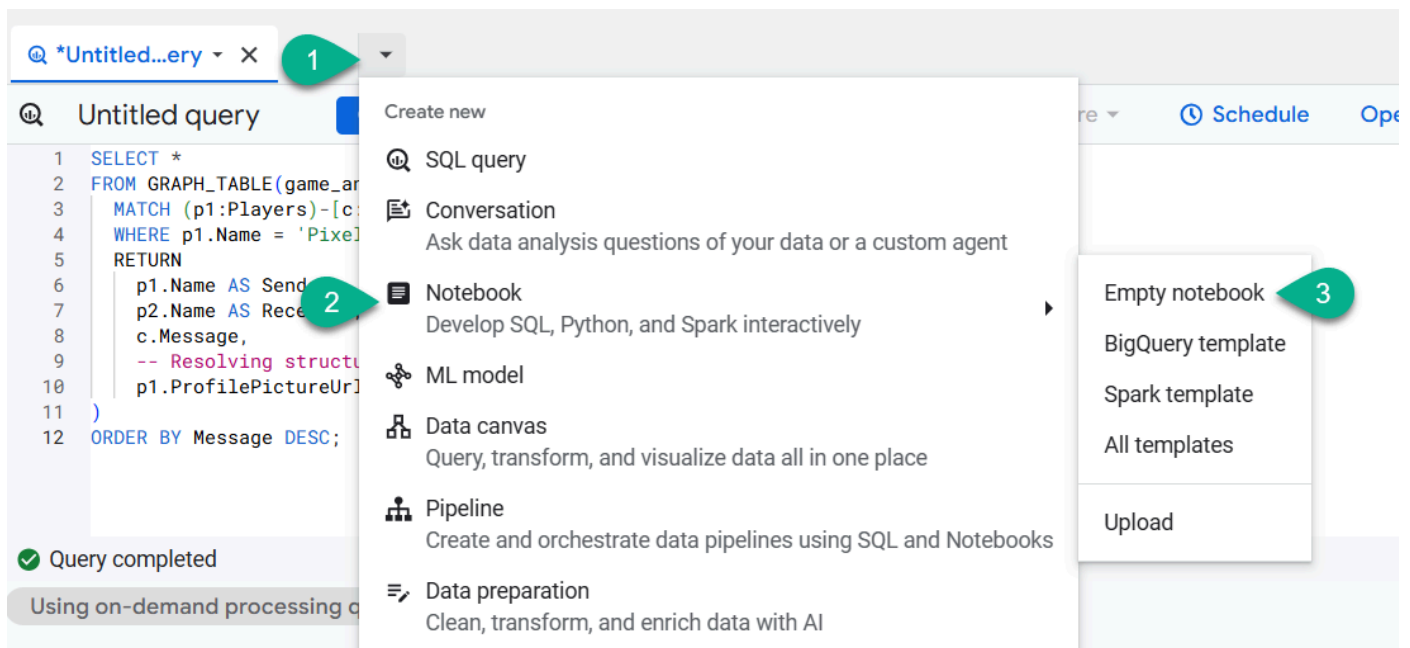
```
-- Compare chat messages between Fraudster A and B
COSINE_DISTANCE(c1.MessageEmbedding, c2.MessageEmbedding) AS MessageDistance,
-- Compare profile pictures between Fraudster A and B
COSINE_DISTANCE(c1.PicEmbedding, c2.PicEmbedding) AS PictureDistance
FROM CatData c1
CROSS JOIN CatData c2
WHERE c1.Sender < c2.Sender -- Avoid self-comparison and duplicate pairs (A-B and B-A)
      AND c1.SenderProfilePic <> c2.SenderProfilePic
ORDER BY PictureDistance ASC, MessageDistance ASC
LIMIT 10;
```

開啟個人資料相片後，你會發現氏族成員的呈現方式十分相似。

以視覺化方式呈現詐欺集團

您可以使用筆記本和 Python Cell Magic，將詐騙集團視覺化。這樣我們就能輕鬆查看圖表結果。如要進一步瞭解這項功能，請參閱[說明文件](#)中的視覺化內容。

在 [BigQuery Studio](#) 中，依序點選「更多」>「Notebook」>「空白 Notebook」。



如果是第一次使用 Notebooks，請按一下「啟用 API」。接著，系統會要求你輸入區域。本實驗室使用 `us`，因此請在 `us` 多區域中選擇一個區域。

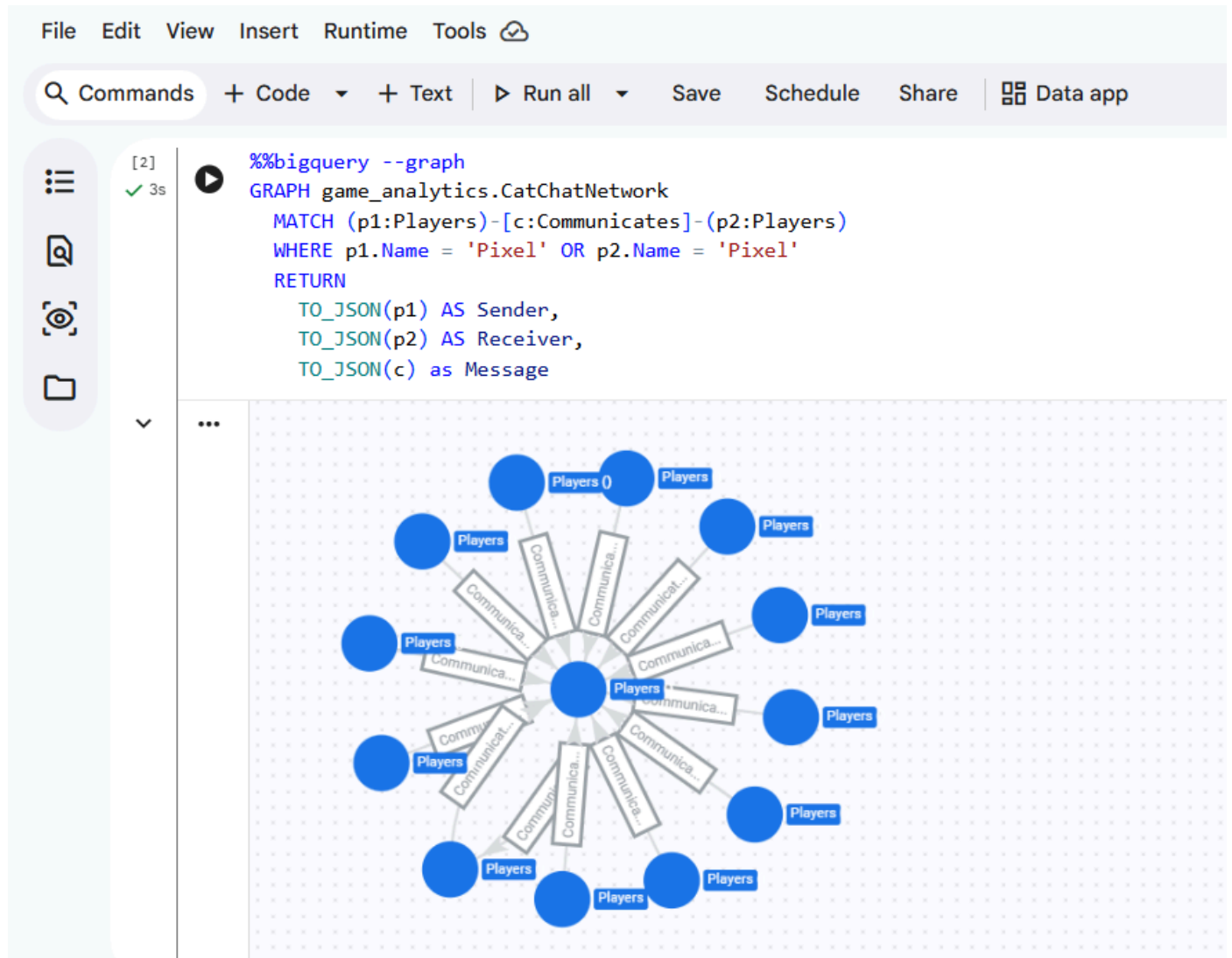
將下列內容貼到程式碼儲存格：

```
!pip install bigquery-magics==0.12.1
```

使用「+ Code」按鈕建立新的儲存格，並貼上下列內容：


```
%%bigquery --graph
GRAPH game_analytics.CatChatNetwork
MATCH p=(p1:Players)-[c:Communicates]-(p2:Players)
WHERE p1.Name = 'Pixel' OR p2.Name = 'Pixel'
RETURN TO_JSON(p) AS full_path
```

按一下「Run all」(全部執行)。大約一分鐘後，您應該會看到通訊網路的圖表。



步驟 7 · 約 5 分鐘

7. 清除

如要避免系統向您的 Google Cloud 帳戶收取本程式碼研究室所用資源的費用，請刪除您建立的資源。

刪除 Spanner 執行個體

```
gcloud spanner instances delete game-instance
```

刪除 BigQuery 資料集

```
bq rm -r -f -d game_analytics
```

或者，刪除專案

如果您是為了這個實驗室建立新專案，可以刪除整個專案：

```
gcloud projects delete <YOUR_PROJECT_ID>
```

步驟 8 · 約 1 分鐘

8. 恭喜！

恭喜！您已成功使用 Spanner 和 BigQuery 建構即時詐欺防禦機制。

您學到了下列內容：

- 使用 **BigQuery 持續查詢**，將即時洞察資料推送至 Spanner。
- 使用 **Spanner Graph** 追蹤財務關係。
- 使用 **Spanner 向量搜尋**，對非結構化資料執行相似度查詢。
- 使用 **BigQuery 圖**追蹤通訊網路。

後續步驟

- 參閱 [Spanner Graph 說明文件](#)。
 - 如需更多程式碼範例，請前往 [Spanner Graph GitHub 存放區](#)。
 - 進一步瞭解 [BigQuery Graph](#)。
 - 請參閱「[BigQuery 持續查詢](#)」。
-